

Till Breuer

✉ breuertill@gmail.com

Mathematician. Data Scientist. Software Developer.

Data Scientist with a math diploma from University of Münster and a background in **machine learning**, **software development** and **statistical modelling**.

EDUCATION

10/2004 - 06/2010

University of Münster

Diploma in Mathematics / Probability Theory Ø 1.2

Thesis (1.3): »Zur Rekurrenz von Random Walks«

10/2001 - 10/2004

Berufsakademie Mannheim

Diploma in Information Technology (BA) Ø 1.5

Thesis (1.4): »Entwurf einer Architektur zur Publikation von Unternehmensverzeichnissen im Intranet«

1991 - 2000

Gymnasium Am Geroweier, Mönchengladbach

Abitur Ø 1.4

EXPERIENCE

Freelance

Self-employed

8/2019 - present

Data Scientist

Projects

As a freelancer, I have tackled algorithmic challenges across various domains, including computer vision, natural language processing, and demand forecasting. In addition to these, I have also engaged in data engineering tasks, focusing on the implementation of data pipelines and the expansion of data warehouses. My role has encompassed the development of concepts, research and development, hands-on implementation, and project coordination tasks.

Projects I have worked on include:

- Swipe-based recommendation app using content-based image retrieval and a bandit algorithm
- Conducting workshops and teaching concepts of deep learning for computer vision
- Text-based categorization of marketing texts
- Hybrid image semantic segmentation and pose detection in street scenes
- Implementation of a literature management system with ELK-stack integration
- NLP- and recommendation-based matchmaking
- Evaluation of deep learning models for autonomous driving
- Time series forecasts for demand prediction in logistics
- Design and implementation of recommendation microservices for production news sites
- Data pipelines and data warehouse design & implementation on AWS and GCloud with dbt and Airflow

Technology stack

- Python (10 years)
- SQL & relational databases (15 years)
- pandas (10 years)
- scikit-learn (10 years)
- PyTorch (5 years)
- matplotlib, altair (10 years)
- Elasticsearch and ELK-stack (5 years)
- GCloud & BigQuery (2 years)
- AWS (1 year)
- dbt (1 year)
- Airflow (1 year)
- Keras + Tensorflow (3 years)
- Ruby (7 years)
- JavaScript, AngularJS (5 years)

mediaire GmbH

AI-software for radiologists

2/2020 - 10/2021

Senior Data Scientist

At mediaire I was responsible for benchmarking and adapting segmentation and classification models for MRI images as well as for integrating them into the software.

Responsibilities & Achievements

- Developing prototypes for segmenting and classifying MRI-images
- Evaluating and benchmarking state-of-the-art solutions for medical image analysis
- Development and improvement of Deep Learning-models

Twenty Billion Neurons GmbH

Realtime video understanding

6/2017 - 8/2019

A.I. Engineer / Senior A.I. Engineer

At Twenty Billion Neurons I worked as a research and development engineer, creating artificial neural networks which detect activities in videos. I participated in developing solutions for spatial and temporal activity detection using deep neural networks and applied the technology in a series of customer projects.

Responsibilities & Achievements

- Developing solutions for spatial detection performed in real-time on videos
- Developing solutions for accurate temporal predictions and prediction of trigger points in videos
- Realization of customer projects
- Mentoring tasks

The unbelievable Machine Company GmbH

Cloud Service Provider and Full Service Provider for Data Solutions

11/2014 - 6/2017

Data Scientist

At the unbelievable Machine Company I worked on a variety of Data Science projects. These include industry projects as well as e-commerce and media-related projects. Among others I worked on

- Computer Vision: Image recognition using neural networks
- Recommendation: Collaborative filtering, content-based recommendations, statistical and bandit approaches
- Predictive maintenance: Predicting malfunction of manufacturing components and classifying errors
- Performance prediction: Predicting distribution of user participation among different channels
- Prediction of energy consumption
- Various clustering / segmentation tasks

Responsibilities & Achievements

- Leading role in most projects
- Customer communication
- Contributing to the improvement of internal processes
- Introducing best practices for coding
- Speaker at several conferences including the bitkom Big Data Summit

bitcrowd GmbH

Web Development Consultancy

8/2014 - 10/2014

Freelance Software Developer

After Krautreporter was crowdfunded I helped developing the platform as leading backend and Javascript engineer. I was responsible for the technical concepts of the platform as well as for implementing the Rails backend, the AngularJS components and making the components work together. The platform had a tight deadline and launched on time.

Responsibilities & Achievements

- Backend and client-side development
- Choices of technology (web content editor for publishing articles)
- Technical concepts

finn GmbH / openproject

Software Development Consultancy

10/2013 - 8/2014

Software Developer

Finnlabs is a Berlin-based company centered around their product, an open source project collaboration software called OpenProject. I joined the company as a senior software developer.

Responsibilities & Achievements

- Backend development (Ruby on Rails)
- Working in an international team
- Requirements engineering and sprint planning
- Team coordination and leading the Javascript / AngularJS development

Zweitag GmbH

Ruby on Rails Consultancy

6/2010 - 10/2013

Software Developer

As a software engineer I worked on a variety of projects, for example so-schmeckts.de or compeon.de. Technologies we worked with involved **Rails** as server-side technology as well as **JavaScript** and related frameworks like **AngularJS** on the client-side. Zweitag applies behaviour driven development using **Rspec**, **Capybara** and **Jasmine**. Data visualization were done with **D3.js** among others.

Responsibilities & Achievements

- Backend and client-side development
- Technical concepts
- Customer communication
- Project coordination

Institute of Mathematical Statistics

of the University of Münster

8/2010 - 10/2011

Research assistant

As a research assistant at the institute for probability theory at the University of Münster I was mentor for seminars in probability theory and statistical modelling. Besides I lead the internship for applied statistics with R. I did research in the field of renewal theory exploring structural inheritance properties of Markov renewal processes. Meanwhile I worked as a software developer at Zweitag GmbH (see above).

Fairix

Startup

10/2008 - 3/2010

Software developer, CTO

Fairix is a startup I took part in as a software engineer while I was a student of mathematics. Fairix provided a platform to producers and merchants enabling them to offer products with additional ecological or social value, e.g. fair trade products. In an engineering team of two we developed a marketplace platform with Ruby on Rails. For that purpose I learnt Rails from scratch and I was leading the development in the later course of the project when the platform launched. The startup finally ran out of funds after two years.

Bundesamt für Sicherheit in der Informationstechnik (BSI)

8/2007 - 9/2007

Research engineer (Pattern recognition)

At my internship at the BSI I worked on research topics about malware classification. The internship lasted two months and together with my mentor I implemented a pattern recognition and a clustering approach to identify and distinguish malware by the kinds of exploits it used. The software had to be extremely performant and of linear complexity and was written in C.

Institute of Mathematical Statistics

of the University of Münster

10/2006 - 3/2008

Tutor in linear algebra / functional analysis

Tutorials and correction of student homework.

INTERESTS

- Music
- Reading
- Studying
- Sports
- Mathematics, Physics, Computer Science, Biology
- Code design

REFERENCES

Stefan Otte, former Lead Data Scientist at paretos GmbH (<https://www.paretos.com/de/home>)

was my team lead at paretos

Nicolas Gorges, Lead Data Scientist at Birds on Mars GmbH (<https://www.birdsonmars.com>)

was my team lead at The unbelievable Machine Company